AMRITA VISHWA VIDYAPEETHAM | School of Engineering

Embedded Systems Project

Ghost Vehicle in Fog Detector Using Pressure Wave Sensing Before Headlights Appear

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Ghost Sense Initiative

The Invisible Threat

- Dense fog critically reduces visual range on highways and rural roads, rendering standard headlights ineffective.
- Drivers face severely delayed detection of approaching vehicles, drastically increasing the likelihood of high-speed, catastrophic collisions.

The Edge-AI Solution

A New Paradigm

The Technology Stack

Zero-Latency Intelligence

SENSE

PROCESS

WARN

Detecting approaching vehicles before they become visible by capturing the low-frequency pressure waves they push ahead of them.

- Utilizes highly sensitive MEMS acoustic sensors. Processes physical data
- locally on a Cortex-M microcontroller.

- Deploys TinyML and Edge-AI directly on the chip. Triggers life-saving early warnings
- without relying on:



SENSE

Capture invisible pressure disturbances generated by moving vehicles.



PROCESS

Execute real-time signal analysis natively on a Cortex-M core.



WARN

Trigger immediate driver alerts through:

- Visual indicators (LEDs/Display)
- Audible alarms (Buzzer)

LITERATURE SURVEY

S.NO	Title of the Paper	Year	Objective	Hardware /Software	Outcome	Inference
1.	**Adaptive Sensor Fusion for Robust Perception in Dense Fog: A Gated Vision and LiDAR Integration Framework**	**JUNE 2026**	Improve obstacle detection in dense fog using adaptive multi-modal sensor fusion.	**Hardware:** Gated Camera, 3D LiDAR. **Software:** AFW-Net, Attention-based Adaptive Fusion, Uncertainty Estimation, Cross-modal Feature Extraction.	Achieved 82% AP at 50 m visibility, improving detection by 23.7% over conventional RGB-LiDAR fusion.	Adaptive sensor weighting, Reliable sensor fusion, Robust fog perception
2.	**Ensemble Artificial Intelligence Framework for Real-Time Automatic Vehicle Detection and Classifications**	**MAY 2026**	Develop an ensemble AI framework for accurate vehicle detection and classification under adverse weather.	**Hardware**: RGB Camera. **Software**: YOLOv5, YOLOv7, YOLOv8, YOLOv10, DETR-ResNet50, Faster R-CNN, Dynamic Weighting, NMS, Ensemble Learning.	Achieved 96.6% mAP, 94.2% Precision, and 94.1% F1-score for multi-class vehicle detection.	Ensemble learning, High detection accuracy, Vision-dependent
3.	**FogGate-YOLO: Traffic Object Detection in Foggy Environments Using Channel Selection Mechanisms**	**MAR 2026**	Design a lightweight object detector specifically optimized for foggy environments.	**Hardware:** RGB Camera. **Software:** YOLOv12, A2C2f-FSA Module, Bidirectional Feature Fusion Module (BFFM), GSConv, F-WIoU v3 Loss.	Improved mAP@0.5 by 5.2% (RTTS) and 7.3% (VOC-FOG) while maintaining real-time inference.	Fog-aware feature enhancement, Real-time detection, Camera limitations
4.	**FogFusion: Robust 3D object detection based on camera-LiDAR fusion for autonomous driving in foggy weather conditions**	**MAR 2026**	Improve 3D object detection in fog using enhanced camera-LiDAR fusion.	**Hardware:** RGB Camera, LiDAR. **Software:** Depth Completion Network with Fog Convolution (DCFC), Flexible Cylindrical Voxel (FCV), Cylindrical Fusion Module (CFM).	Improved detection performance by ≥3.32% over baseline methods on KITTI-C.	Enhanced 3D perception, Virtual point clouds, Costly sensor fusion
5.	**Dynamic Visibility Recognition and Driving Risk Assessment Under Rain–Fog Conditions Using Monocular Surveillance Imagery**	**JAN 2026**	Estimate road visibility and driving risk under rain-fog conditions using roadside surveillance images.	**Hardware:** Monocular CCTV Camera. **Software:** Koschmieder Scattering Model, Dark Channel Prior, Monte Carlo Simulation, Low Visibility Risk Field Force (LVRFF).	Achieved >86% visibility estimation accuracy and effectively identified high-risk driving zones.	Visibility estimation, Risk assessment, Infrastructure-based monitoring

OBJECTIVES

- To detect approaching vehicles before their headlights become visible by sensing the low-frequency pressure waves they push ahead of them, using a MEMS sensor.
- To condition the sensor signal (amplification and band-pass filtering with an LM358 op-amp) so that only the vehicle-related frequencies remain.
- To acquire the signal through the ADC of a Cortex-M microcontroller and clean it with digital filtering (moving average).
- To warn the driver with immediate visual alerts (LEDs, LCD/OLED display) and an audible alert (buzzer) before the vehicle becomes visible.
- To collect data from live moving vehicles, then test, calibrate and optimise the system for reliable detection.
- To build a complete working prototype and evaluate its performance, aiming to improve road safety in low-visibility conditions.

Architecture & Data Pipeline

From physical pressure waves to real - time Edge-AI alerts.

Zone 1
Physical Sensing

- Vehicle Approaches
- Acoustic Detection
- Signal Conditioning

Zone 2
Edge-AI Processing

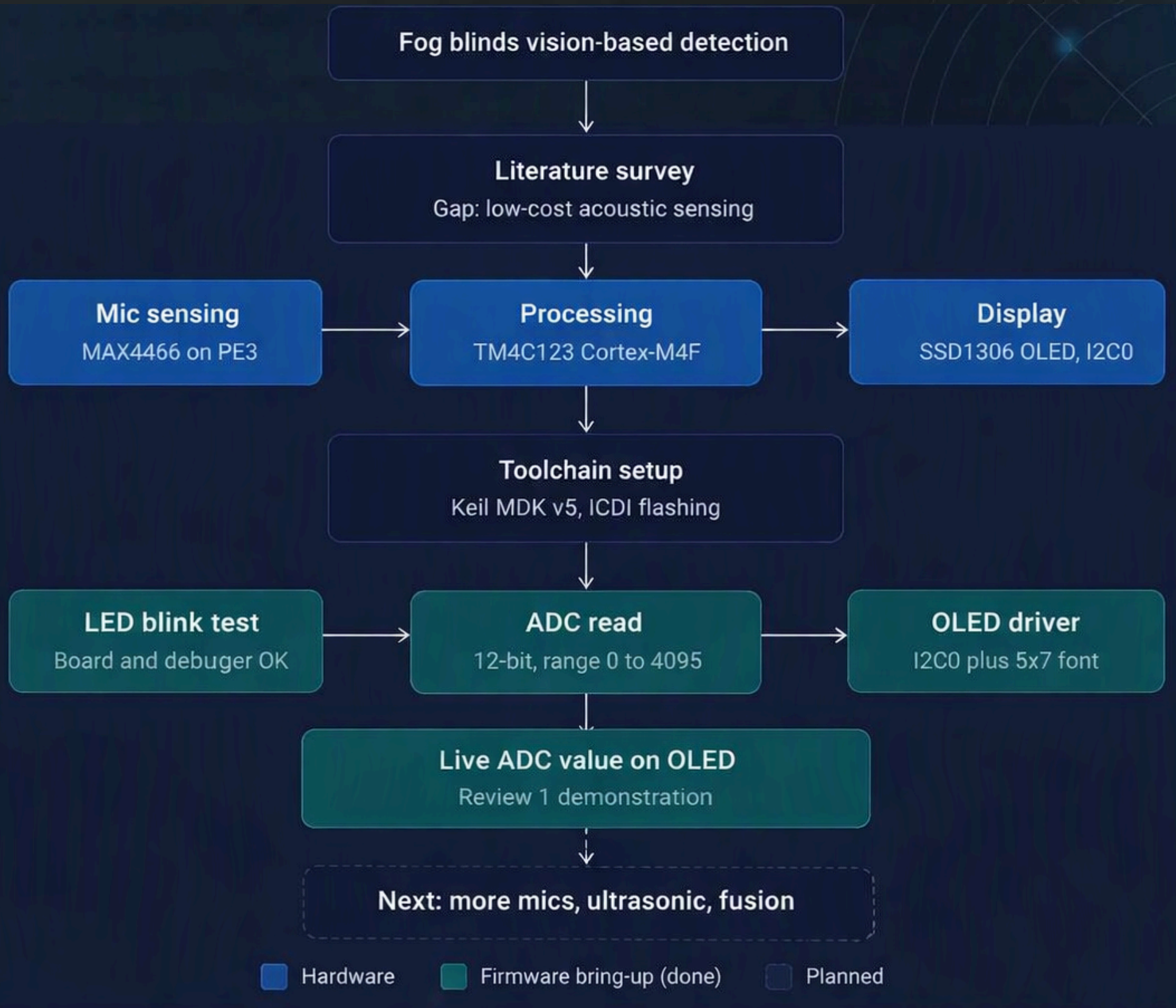
- 4. Data Acquisition
- 5. Digital Filtering
- 6. Algorithmic Analysis
- 7. Threshold Decision

Zone 3
Alert Execution

- 8. Confirmation
- 9. Early Warning
- 10. Result

Pressure Waves > MEMS Sensor > Signal Conditioning > Cortex-M ADC > Digital Filtering > TinyML Analysis > Vehicle Detection > LED/LCD/Buzzer Alert > Driver Safety

METHODOLOGY



Phase & Focus	Week	Core Activity	Key Milestone
Phase 1: Foundation(Research & Hardware)	Week 1-3	Literature Survey, Problem Analysis & Component Procurement.STM32/Cortex-M setup and architecture learning.	Hardware Procured & Environment Setup
Phase 2: Integration(Interfacing & Data)	Week 4 - 6	Sensor Interfacing & ADC Testing.Data Collection from live moving vehicles.Initial Signal Processing Implementation.	Successful Sensor Data Acquisition
Phase 3: Edge-AI(Algorithm & Alert)	Week 7 - 9	Vehicle Detection Algorithm (TinyML) Development.LED, Buzzer, and Display Integration.System Testing, Calibration, and Optimisation.	Functional Detection Algorithm
Phase 4: Finalisation(Prototype & Delivery)	Week 10 - 12	Complete Prototype Assembly.Result Analysis and Performance Evaluation.Report Writing and PPT Preparation.	Final Demonstration (Sept 20)

Hardware Ecosystem

Compute
- Tiva C Board
- Cortex-M Microcontroller

Sensing
- MEMS Microphone Sensor
- MAX9814 / MAX4466

Conditioning
- LM358 Operational Amplifier
- Used for signal amplification

Output UI
- OLED Display / LCD (16×2)
- Warning LEDs
- Buzzer

Ancillary Components
- USB Programmer / ST-Link
- Breadboard
- Power Supply

Software Ecosystem

Environment
- Keil uVision5

Programming Language
- Embedded C Programming
- Optimized for TinyML constraints

Analysis Tools
- MATLAB
- Python
- Used for initial signal analysis and algorithm modeling

Simulation & Debugging
- Serial Terminal
  - PuTTY
  - Tera Term
- Proteus (optional for simulation)

Expected Output

Early Detection : Reliable identification of approaching vehicles completely obscured by dense fog.

Immediate Alert : Distinct visual and audible warnings delivered to the driver via the edge device with zero latency.

Ultimate Impact : Significantly improved road safety and a drastic reduction in collisions under critical low-visibility conditions.

Thank You!